

3	(a) the point where (all) the weight (of the body) is considered / seems to act	M1 A1	[2]
(b) (i)	vertical component of $T (= 30 \cos 40^\circ) = 23 \text{ N}$	A1	[1]
(ii)	the <u>sum</u> of the clockwise moments about a <u>point</u> equals the <u>sum</u> of the anticlockwise moments (about the same point)	B1	[1]
(iii)	(moments about A): 23×1.2 (27.58) $= 8.5 \times 0.60 + 1.2 \times W$ working to show $W = 19$ or answer of 18.73 (N)	M1 M1 A1	[3]
(iv)	$(M = W / g = 18.73 / 9.81 =) 1.9(09) \text{ kg}$	A1	[1]

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- (c) (for equilibrium) resultant force (and moment) = 0 B1
- upward force does not equal downward force / horizontal component of T
- not balanced by forces shown B1 [2]